

Feature Selection And Classification Accuracy On High Dimensional Gene Expression Data

Comprehensive Literature Review and Methodology

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Introduction

The High-Dimensional Challenge i

- Gene expression data derived from high-throughput technologies such as RNA sequencing have become an important resource for studying biological processes and disease, and also generate extremely large biological datasets.
- As a result, these datasets are characterized by the **high-dimensional, low-sample-size (HDLSS)** property, where the number of features greatly exceeds the number of observations.
- In most biomedical studies the number of variables greatly exceeds the number of observations. Formally, if a dataset is represented by a matrix $X \in R^{n \times p}$, where n denotes the number of samples and p denotes the number of features (genes) which often satisfy (Shroff and Maheta [2015])

$$p \gg n$$

The High-Dimensional Challenge ii

- Because many genes may be irrelevant or redundant for a given classification task, directly applying machine learning algorithms to such high-dimensional data may lead to poor predictive performance.(noa [2018])
 - **Curse of Dimensionality:** As dimensionality increases, the data space becomes sparse and the distance between points loses meaning, making statistical learning and pattern recognition significantly more difficult (Venkatesh and Anuradha [2019]).
 - **Redundancy:** Many genes are highly correlated because they belong to the same biological pathway. Including all of them introduces redundant information that increases computational cost without improving predictive ability.
 - **Irrelevance:** A large portion of genes are not associated with the disease phenotype being studied. These irrelevant variables can introduce noise and degrade classifier performance (Kumar and Minz [2014]).

Objective: Identify a smaller subset, of genes $S \subseteq \{1, \dots, p\}$ that preserves the biological signal while improving classification accuracy and interpretability.

The field of gene expression analysis began to expand rapidly with the introduction of high-throughput genomic technologies in the late 1990s.

- **Golub et al. (1999):** One of the earliest case studies demonstrating that gene expression profiles could be used for molecular classification of cancers. Their work successfully distinguished Acute Lymphoblastic Leukemia (ALL) from Acute Myeloid Leukemia (AML), establishing the foundation of genomic classification.(noa)
- **Guyon et al. (2002):** Proposed **Recursive Feature Elimination (RFE)**, a powerful wrapper-based feature selection technique. The method iteratively removes genes with the smallest contribution to the classifier by examining SVM weight magnitudes (Guyon et al. [2002]).

- **Bellman's Legacy:** Richard Bellman's concept of the *Curse of Dimensionality* highlighted the difficulty of searching extremely large feature spaces. With p genes, the number of possible subsets equals 2^p , making exhaustive search computationally infeasible for genomic datasets.

These developments motivated the creation of efficient feature selection strategies specifically designed for high-dimensional biological data.

Methodology & Framework

The Feature Selection Pipeline i

Feature selection algorithms typically follow a structured pipeline consisting of four major stages (Kumar and Minz [2014]):

1. **Subset Generation** A search procedure generates candidate subsets of features from the original set of genes.
2. **Subset Evaluation** Each candidate subset is evaluated using a statistical measure or classification performance metric.
3. **Stopping Criterion** The search terminates when a predefined condition is satisfied, such as reaching a fixed number of features or when performance stops improving.
4. **Validation** The selected feature subset is tested on independent or cross-validation datasets to ensure generalization.

This framework ensures that the resulting feature set balances predictive accuracy with computational efficiency.

Finding the optimal subset of genes is an NP-hard problem due to the exponential growth of possible combinations.

Several search strategies have been proposed to address this challenge:

- **Sequential Search Methods** These include Sequential Forward Selection (SFS) and Sequential Backward Selection (SBS). SFS begins with an empty set and adds features iteratively, whereas SBS starts with the full feature set and removes the least useful genes (Tang et al. [2013]).
- **Heuristic Search Methods** Evolutionary algorithms such as Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) explore the feature space using population-based strategies that mimic biological evolution or swarm intelligence.

- **Randomized Search Methods** Algorithms such as Las Vegas Filter (LVF) perform stochastic exploration of the search space, selecting subsets randomly and retaining those that satisfy predefined evaluation criteria.

These strategies aim to efficiently navigate the extremely large search space while maintaining acceptable computational complexity.

Taxonomy of Models

Filter Methods: Statistical Efficiency i

Filter methods evaluate the relevance of genes using statistical properties of the data, independent of any classification algorithm (Ferreira and Figueiredo [2012]).

Because they do not rely on training a predictive model, they are computationally efficient and scalable to very high-dimensional datasets.

Common filter techniques include:

- **Information Gain (IG)** Measures how much information a gene contributes toward predicting the class label by calculating entropy reduction.
- **Fisher Score** Ranks genes based on the ratio between inter-class variance and intra-class variance, identifying genes that best separate different disease classes.

- **ReliefF Algorithm** A distance-based method that estimates feature importance by comparing neighboring samples from the same and different classes (Li et al. [2017]).

Advantages:

- Fast computation
- Scalability to tens of thousands of genes
- Reduced risk of overfitting

Wrapper Models

Wrapper methods evaluate feature subsets by directly measuring classifier performance. A learning algorithm such as Support Vector Machines or k-Nearest Neighbors is treated as a black-box evaluator.

Although wrappers often achieve higher predictive accuracy, they are computationally expensive when dealing with thousands of genes (Tang et al. [2013]).

Embedded Models

Embedded methods integrate feature selection directly into the model training process.

- **LASSO (L_1 Regularization)** Adds a penalty term that forces many coefficients to shrink to zero, effectively performing feature selection during optimization.
- **Random Forests** Provide measures such as Gini importance or permutation importance that rank genes according to their contribution to predictive accuracy.

Embedded approaches often provide a balance between computational efficiency and predictive performance.

Modern Data Perspectives

Advanced Challenges: Streaming & Structure i

Recent developments in genomics introduce new complexities beyond traditional static datasets Li et al. [2017].

- **Streaming Features** In modern sequencing pipelines, gene features may appear sequentially over time. Online feature selection algorithms must dynamically update the selected subset as new data arrives.
- **Structured Features** Genes are often organized into biological pathways or regulatory networks. Techniques such as Group LASSO incorporate this structure by selecting entire gene groups rather than individual genes (Tang et al. [2013]).
- **Multi-view Data Integration** Modern biomedical datasets combine multiple data modalities including RNA-seq, DNA methylation, and clinical attributes. Feature selection methods must therefore integrate heterogeneous data sources.

These challenges require more sophisticated feature selection models capable of capturing biological relationships between genes.

Analysis & Synthesis

Mathematical Formulation i

Feature selection can be formulated as an optimization problem where we seek a sparse indicator vector

$$w \in \{0, 1\}^p$$

that determines which genes are selected.

$$\min_w \text{Loss}(f(X \odot w, y)) + \lambda \|w\|_0$$

- $X \odot w$ represents the dataset restricted to the selected genes.
- $\|w\|_0$ denotes the L_0 norm, corresponding to the number of non-zero entries in w , or equivalently the number of selected genes.
- λ controls the trade-off between model complexity and predictive performance (Shroff and Maheta [2015]).

The goal is therefore to identify a small subset of genes that minimizes prediction error while maintaining model sparsity.

Several recent studies highlight the importance of appropriate feature selection in genomic classification.

- **Källberg (2021):** Demonstrated that carefully selected gene subsets significantly improve subtype identification in RNA-seq datasets compared to naive high-variance filtering approaches (Källberg et al. [2021]).
- **García-Díaz (2020):** Reported classification accuracies exceeding 98% using unsupervised Genetic Algorithm-based feature selection for multi-class cancer datasets (García-Díaz et al. [2020]).

These results confirm that intelligent feature reduction is a critical component of high-dimensional biomedical data analysis.

Conclusion and Research Gap i

Despite decades of research, there is still no universally optimal feature selection technique for high-dimensional genomic data.

- Filter methods provide efficient initial screening of genes.
- Wrapper and embedded methods often yield higher predictive accuracy but at higher computational cost.

Research Gap:

Current literature lacks consensus on hybrid frameworks that effectively combine the speed of filter methods with the accuracy of wrapper or embedded approaches, particularly for extremely sparse RNA-seq datasets.

This motivates our study to systematically evaluate different feature selection strategies and assess their impact on classification accuracy.

References

(PDF) Molecular classification of cancer: class discovery and class prediction by gene monitoring. URL https://www.researchgate.net/publication/12779876_Molecular_classification_of_cancer_class_discovery_and_class_prediction_by_gene_monitoring.

[//www.researchgate.net/publication/12779876_Molecular_classification_of_cancer_class_discovery_and_class_prediction_by_gene_monitoring](https://www.researchgate.net/publication/12779876_Molecular_classification_of_cancer_class_discovery_and_class_prediction_by_gene_monitoring).

High dimensional data classification and feature selection using support vector machines. *European Journal of Operational Research*, 265(3):993–1004, Mar. 2018.

ISSN 0377-2217. doi: 10.1016/j.ejor.2017.08.040. URL <https://www.sciencedirect.com/science/article/abs/pii/S0377221717307713>.

[//www.sciencedirect.com/science/article/abs/pii/S0377221717307713](https://www.sciencedirect.com/science/article/abs/pii/S0377221717307713).

A. J. Ferreira and M. A. T. Figueiredo. Efficient feature selection filters for high-dimensional data. *Pattern Recognition Letters*, 33:1794–1804, 2012.

P. García-Díaz, I. Sánchez-Berriel, J. A. Martínez-Rojas, and A. M. Díez-Pascual. Unsupervised feature selection algorithm for multiclass cancer classification of gene expression RNA-Seq data. *Genomics*, 112:1916–1925, 2020.

- I. Guyon, J. Weston, S. Barnhill, and V. Vapnik. Gene Selection for Cancer Classification using Support Vector Machines. *Machine Learning*, 46:389–422, 2002.
- V. Kumar and S. Minz. Feature selection: A literature review. *Smart Computing Review*, 4(3):211–229, 2014.
- D. Källberg, L. Vidman, and P. Rydén. Comparison of Methods for Feature Selection in Clustering of High-Dimensional RNA-Sequencing Data to Identify Cancer Subtypes. *Front. Genet.*, 12:632620, 2021.
- J. Li, K. Cheng, S. Wang, F. Morstatter, R. P. Trevino, J. Tang, and H. Liu. Feature selection: A data perspective. *ACM Computing Surveys (CSUR)*, 50(6), 2017. doi: 10.1145/3136625.
- K. P. Shroff and H. H. Maheta. A comparative study of various feature selection techniques in high-dimensional data set to improve classification accuracy. In *2015 International Conference on Computer Communication and Informatics (ICCCI)*, pages 1–6, 2015.
- J. Tang, S. Alelyani, and H. Liu. Feature selection for classification: A review. In *Data Classification: Algorithms and Applications*. CRC Press, 2013.

- B. Venkatesh and J. Anuradha. A review of feature selection and its methods. *Cybernetics and Information Technologies*, 19(1):3–26, 2019. doi: 10.2478/cait-2019-0001.